

Machine Learning

Introduction - Week 1

Objectives

- Types of Data
- Supervised Learning Key Concepts:
 - Labeled Data
 - Classification vs Regression
- ML Cycle

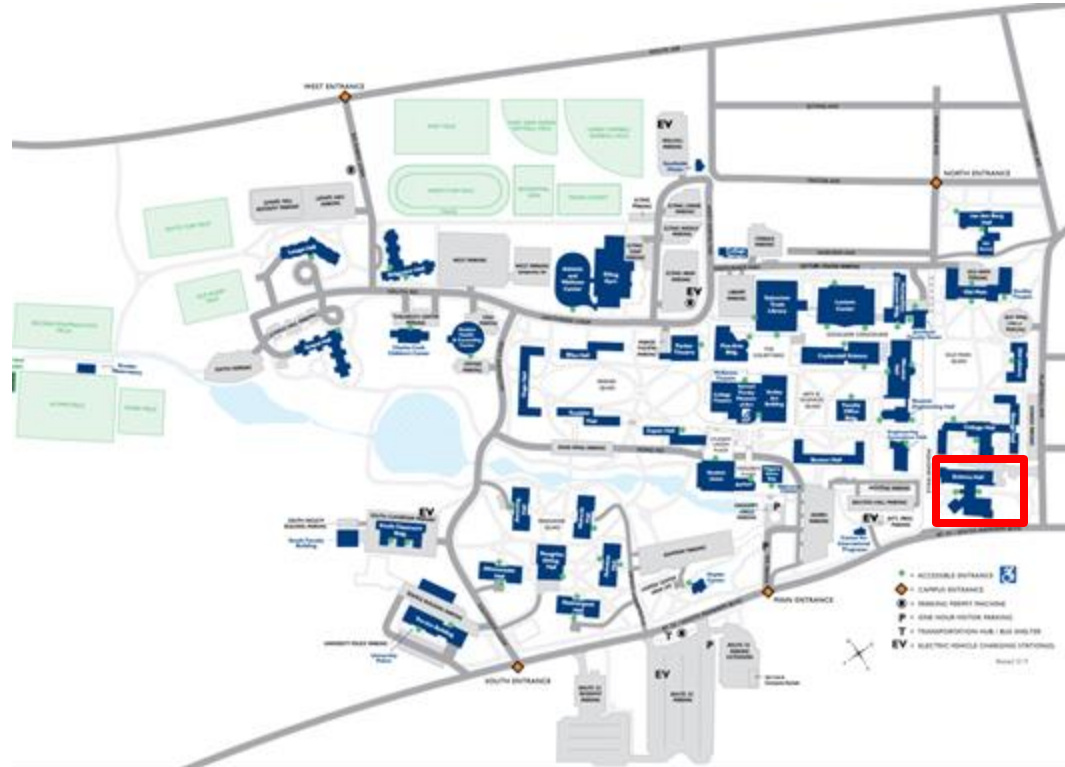
Where to find me?

Office:

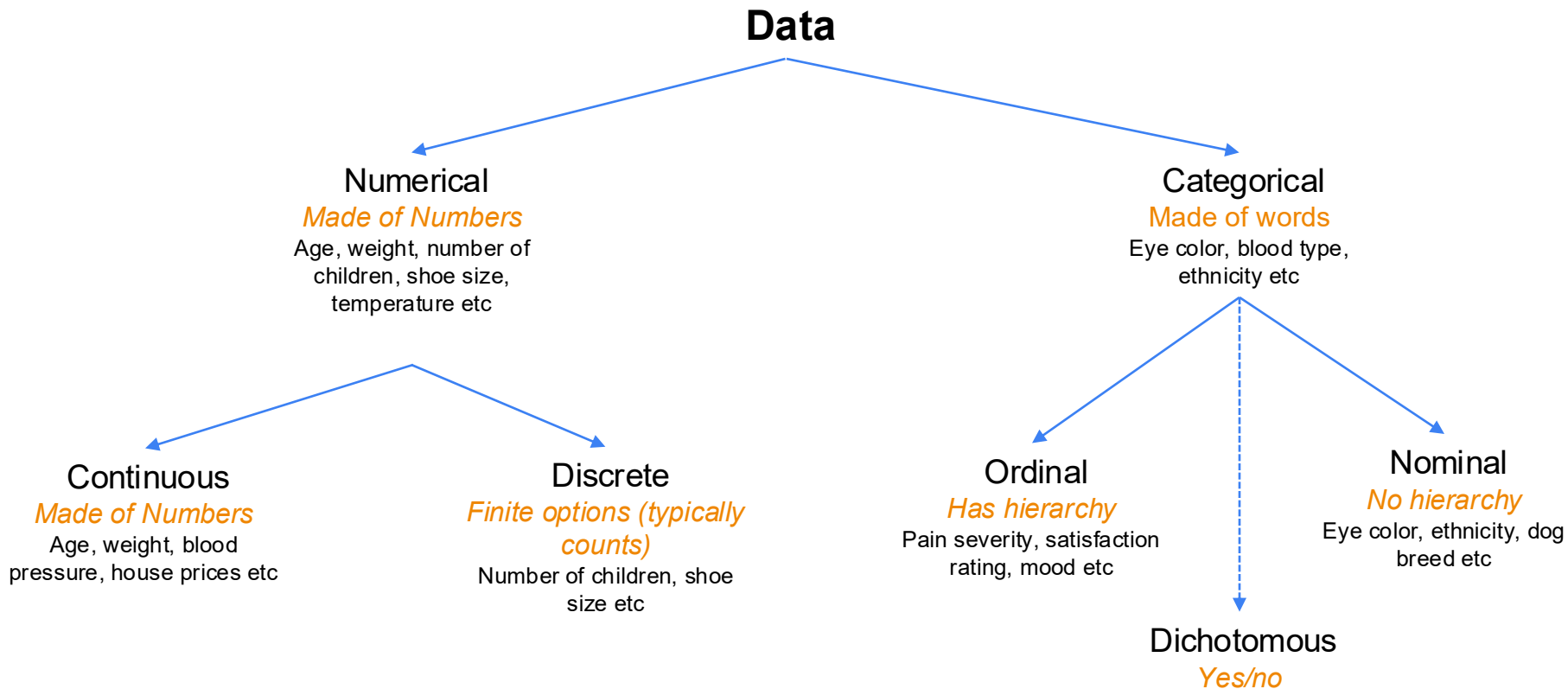
Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM -
1:00 PM EST or by
appointment**



Types of Data



Let's look into the Housing prices

Given, some data on housing prices in rural Indiana, our **task** is to find the price of the a house with the following details:

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				

Table: Inventory of houses sold in Rural Indiana

Supervised Learning

This is an example of **supervised learning** as we are trying to map the input features (total area, # bedrooms, # bathrooms, has finished basement) to the output/target (sales price).

What we end up getting almost all the time is **an approximation of the actual sales price**.

Features				Target
Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				
Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

Mathematically, if X represents the features and Y our target variable, through supervised learning, we estimate the function f , which is given by:

$$\hat{Y} = f(X)$$

Note, $\hat{Y} \neq Y$, meaning the actual house price might be different than what we calculate.

To evaluate our model, we use another function:
 $\mathcal{L}(\hat{Y}, Y)$

Supervised Learning

The supervision in supervised learning is coming from **labeled data**, i.e. **input-output pairs** that is helping the model to learn.

Unsupervised learning on the other hand deals with **unlabeled data**.

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				

Available Data (or Training Data)

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

New Data (or Test Data)

Note that the model, we create is very specific to the type of problem and environment from which the data comes from.

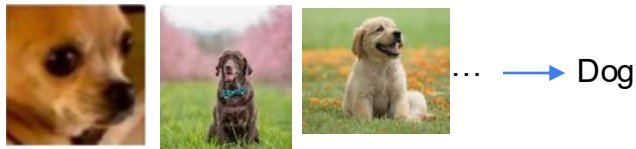
I.e. The sales price of houses in NYC is going to be very different from rural Indiana. The same model would not give us accurate sales prices of houses in NYC. The data from rural Indiana would be considered **out-of-distribution** for NYC.

Note, that the concept of training, test and out-of-distribution data is also present in unsupervised learning.

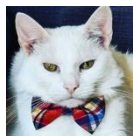
Pop Quiz

You are tasked with distinguishing images of cats from dogs with ML. You have a big dataset with over 400 images of cats and dogs as shown below. Is this task an example of :

- Supervised Learning
- Unsupervised Learning



Training Data



→ Cat/Dog?

Test Data

Pop Quiz

You have a big dataset with over 400 images of cats and dogs as shown below. Your friend gives you a photo of an animal he claims to be a dog but you are not sure, therefore you create an ML algorithm. Is this task an example of :

- Supervised Learning
- Unsupervised Learning



Training Data



Test Data

Classification

vs

Regression

In supervised learning, when the target variable is a **discrete category**

Example: spam/not spam, cat/dog/horse/fish , recognizing handwritten digits

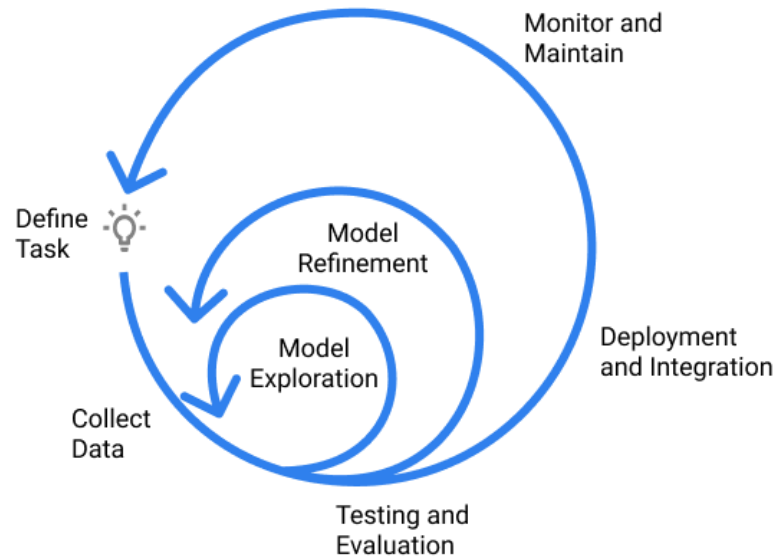
In supervised learning, when the target variable is **continuous**

Example: predicting house prices, stock prices, temperatures etc.

Can you think of a way to convert continuous variable/data to a discrete category?

ML Development Cycle

- Task/Problem definition
- Data Collection
 - Data cleaning
 - Data analysis
- Model exploration and refinement
- Deployment and integration
- Maintenance



ANY QUESTIONS?